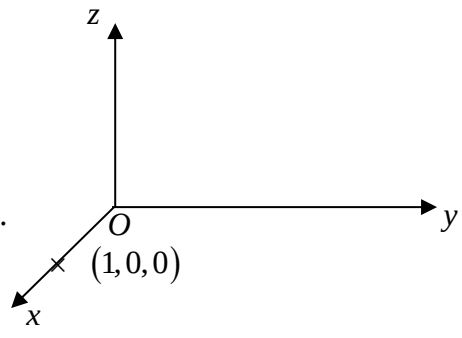
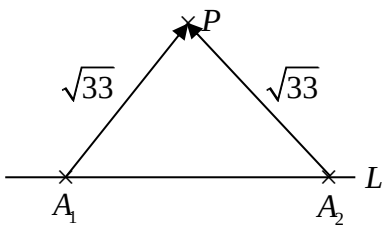
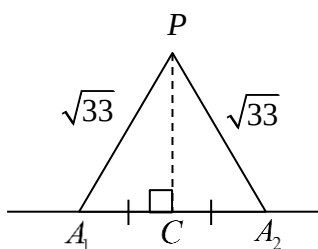
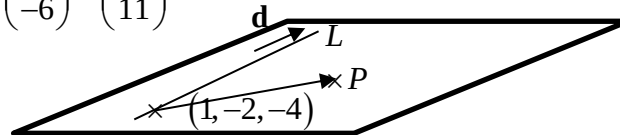
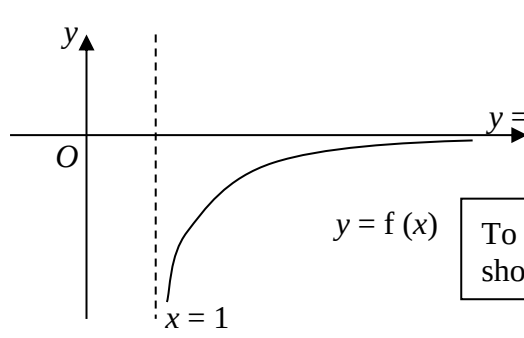


2015 A Level paper 2 Suggested Solutions

1	Suggested Solutions
(i)	For maximum h, $\frac{dh}{dt} = 0$ $\Rightarrow \frac{1}{10} \sqrt{16 - \frac{h}{2}} = 0$ $16 - \frac{h}{2} = 0$ $h = 32 \text{ m}$
(ii)	$\frac{dh}{dt} = \frac{1}{10} \sqrt{16 - \frac{h}{2}}$ $\frac{dh}{dt} = \frac{1}{10} \sqrt{\frac{32 - h}{2}}$ $\int \frac{1}{\sqrt{32 - h}} dh = \int \frac{1}{10\sqrt{2}} dt$ $\int (32 - h)^{-\frac{1}{2}} dh = \int \frac{1}{10\sqrt{2}} dt$ $\frac{(32 - h)^{\frac{1}{2}}}{\frac{1}{2}(-1)} = \frac{t}{10\sqrt{2}} + C$ $-2\sqrt{32 - h} = \frac{t}{10\sqrt{2}} + C$ Since $h = 0, t = 0 \Rightarrow -2\sqrt{32} = C \Rightarrow C = -8\sqrt{2}$ $\therefore -2\sqrt{32 - h} = \frac{t}{10\sqrt{2}} - 8\sqrt{2}$ $t = 10\sqrt{2} (8\sqrt{2} - 2\sqrt{32 - h})$ To reach $h = 16 \text{ m}$, Half of maximum height (32 m) $-2\sqrt{16} = \frac{t}{10\sqrt{2}} - 8\sqrt{2}$ $t = 160 - 80\sqrt{2}$ $\approx 46.9 \text{ years (correct to 3 s.f.)}$ Recall $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ Replacing x by $ax + b$, $\int (ax + b)^n dx = \left(\frac{1}{a}\right) \frac{(ax + b)^{n+1}}{n+1} + C$

Qn 2	Suggested Solutions
(i)	$r = \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}, \lambda \in \mathbb{R}$ A direction vector parallel to x-axis = $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.  $\cos \theta = \frac{\left \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right }{\sqrt{2^2 + 3^2 + (-6)^2} \sqrt{1^2 + 0^2 + 0^2}}$ $\theta = \cos^{-1}\left(\frac{2}{7}\right) = 73.398^\circ \approx 73.4^\circ \text{ (correct to 1 d.p.)}$
(ii)	$\overrightarrow{OP} = \begin{pmatrix} 2 \\ 5 \\ -6 \end{pmatrix}$ Let A be the point on L that is $\sqrt{33}$ units from P. $\overrightarrow{OA} = \begin{pmatrix} 1+2\lambda \\ -2+3\lambda \\ -4-6\lambda \end{pmatrix}, \text{ for some } \lambda \in \mathbb{R}.$ $\overrightarrow{PA} = \begin{pmatrix} 1+2\lambda \\ -2+3\lambda \\ -4-6\lambda \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \\ -6 \end{pmatrix} = \begin{pmatrix} -1+2\lambda \\ -7+3\lambda \\ 2-6\lambda \end{pmatrix}$ $ \overrightarrow{PA} = \sqrt{(-1+2\lambda)^2 + (-7+3\lambda)^2 + (2-6\lambda)^2} = \sqrt{33}$ $4\lambda^2 - 4\lambda + 1 + 9\lambda^2 - 42\lambda + 49 + 4 - 24\lambda + 36\lambda^2 = 33$ $49\lambda^2 - 70\lambda + 21 = 0$ $7\lambda^2 - 10\lambda + 3 = 0$ $(7\lambda - 3)(\lambda - 1) = 0$ $\therefore \lambda = \frac{3}{7} \text{ or } \lambda = 1.$ $\therefore \overrightarrow{OA} = \begin{pmatrix} 3 \\ 1 \\ -10 \end{pmatrix} \text{ or } \frac{1}{7} \begin{pmatrix} 13 \\ -5 \\ -46 \end{pmatrix}$ $\Rightarrow \text{coordinates of A is } (3, 1, -10) \text{ or } \left(\frac{13}{7}, -\frac{5}{7}, -\frac{46}{7}\right).$  There should be 2 points on L that is $\sqrt{33}$ from P $\Rightarrow$ 2 values of λ Need to express as coordinates !

	Let C be the point on L that is nearest to P. By ratio theorem, $\overrightarrow{OC} = \frac{\overrightarrow{OA_1} + \overrightarrow{OA_2}}{2}$ $= \frac{\begin{pmatrix} 3 \\ 1 \\ -10 \end{pmatrix} + \frac{1}{7} \begin{pmatrix} 13 \\ -5 \\ -46 \end{pmatrix}}{2} = \frac{1}{7} \begin{pmatrix} 17 \\ 1 \\ -58 \end{pmatrix} \therefore \text{coordinates of } C \text{ is } \left(\frac{17}{7}, \frac{1}{7}, -\frac{58}{7} \right).$ 
(iii)	A direction vector parallel to plane = $\begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \\ -6 \end{pmatrix} = \begin{pmatrix} -1 \\ -7 \\ 2 \end{pmatrix}$. $\therefore$ normal vector to plane = $\begin{pmatrix} -1 \\ -7 \\ 2 \end{pmatrix} \times \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix} = \begin{pmatrix} 36 \\ -2 \\ 11 \end{pmatrix}$ $\mathbf{r} \cdot \begin{pmatrix} 36 \\ -2 \\ 11 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 36 \\ -2 \\ 11 \end{pmatrix} = -4$ $\therefore \mathbf{r} \cdot \begin{pmatrix} 36 \\ -2 \\ 11 \end{pmatrix} = -4$ $\Rightarrow 36x - 2y + 11z = -4.$  Find 2 vectors parallel to plane and do cross product to obtain normal vector of plane.

Qn 3	Suggested Solutions
(a)(i)	 To prove f^{-1} exist $\Leftrightarrow$ show f is one-one (horizontal line test) Since any horizontal line $y = a$ ($a \in \mathbb{R}$) cuts the graph of f at most once, therefore f is one-one. Hence, f has an inverse. (shown)

(a)(ii)	Let $y = \frac{1}{1-x^2}$ $1-x^2 = \frac{1}{y}$ $x^2 = 1 - \frac{1}{y}$ $x = \pm \sqrt{1 - \frac{1}{y}}$ $\because x > 1, x = \sqrt{1 - \frac{1}{y}}$ $\therefore f^{-1}(x) = \sqrt{1 - \frac{1}{x}}$ Find R_f by sketching graph of f and find range of possible y values. (i.e. minimum and maximum y values) $D_{f^{-1}} = R_f = (-\infty, 0)$
(b)	Let $y = k$ $k = \frac{2+x}{1-x^2}$ $k - kx^2 = 2 + x$ $kx^2 + x + 2 - k = 0$ Find range of values of k for which $y = k$ (horizontal line) and $y = \frac{2+x}{1-x^2}$ intersects. If both graph intersects $\Rightarrow y = k$ is part of range of g $\therefore$ Discriminant ≥ 0 Intersects $\Rightarrow$ real roots (Discriminant ≥ 0) $1^2 - 4(k)(2-k) \geq 0$ $4k^2 - 8k + 1 \geq 0$ $\Rightarrow \left[k - \left(1 + \frac{\sqrt{3}}{2} \right) \right] \left[k - \left(1 - \frac{\sqrt{3}}{2} \right) \right] \geq 0$ $\begin{array}{ccccccc} & + & & - & & + & \\ & \bullet & & \bullet & & & \\ & 1 - \frac{\sqrt{3}}{2} & & 1 + \frac{\sqrt{3}}{2} & & & \end{array}$ $\therefore k \leq 1 - \frac{\sqrt{3}}{2}$ or $k \geq 1 + \frac{\sqrt{3}}{2}$. $\therefore R_g = \left(-\infty, 1 - \frac{\sqrt{3}}{2} \right] \cup \left[1 + \frac{\sqrt{3}}{2}, \infty \right)$ Consider $4k^2 - 8k + 1 = 0$ $\therefore k = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(4)(1)}}{2(4)}$ $= \frac{8 \pm \sqrt{48}}{8}$ $= 1 \pm \frac{\sqrt{3}}{2}.$ Alternatively, we can complete the square.

Qn 4	Suggested Solutions
(a)	[Out of syllabus (9758)]
(b)(i)	$\frac{2}{4r^2 + 8r + 3} = \frac{2}{(2r+1)(2r+3)}$ $= \frac{A}{2r+1} + \frac{B}{2r+3}$ Alternative: Cover-up method $= \frac{A(2r+3) + B(2r+1)}{4r^2 + 8r + 3}$ $\therefore 2 = A(2r+3) + B(2r+1)$ for all real values of r Substitute $r = -\frac{1}{2}$, $A(-1+3) = 2$ Substitute $r = 0$, $3(1) + B = 2$ $A = 1$ $B = -1$ $\therefore \frac{2}{4r^2 + 8r + 3} = \frac{1}{2r+1} - \frac{1}{2r+3}$
(b)(ii)	$S_n = \sum_{r=1}^n \frac{2}{4r^2 + 8r + 3}$ $= \sum_{r=1}^n \left(\frac{1}{2r+1} - \frac{1}{2r+3} \right)$ Note that this is in the form $\sum [f(r) - f(r+1)]$ (MOD) $= \frac{1}{3} - \frac{1}{5}$ $+ \frac{1}{5} - \frac{1}{7}$ $+ \frac{1}{7} - \frac{1}{9}$ $+ \frac{1}{9} - \frac{1}{11}$ $+ \dots$ $+ \frac{1}{2n+1} - \frac{1}{2n+3}$ $= \frac{1}{3} - \frac{1}{2n+3}$ - List downwards - show 3 cancellations - Number of terms left at the start = number of terms left at the end.
(b)(iii)	As $n \rightarrow \infty$, $\frac{1}{3} - \frac{1}{2n+3} \rightarrow \frac{1}{3}$ Sum to infinity $\left \frac{1}{3} - S_n \right \leq 10^{-3}$ Difference between "Sum to infinity" and S_n is within (less than or equal) 10^{-3}. $\left \frac{1}{3} - \left(\frac{1}{3} - \frac{1}{2n+3} \right) \right \leq 10^{-3}$ $\left \frac{1}{2n+3} \right \leq 10^{-3}$ $\frac{1}{2n+3} \leq 0.001, \quad n \geq 0, \quad n \in \mathbb{Z}^+$

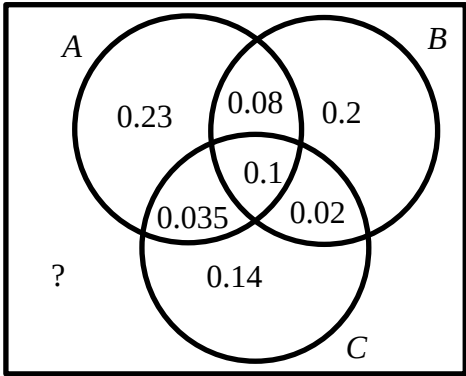
	Using GC, $n = 498, \frac{1}{2n+3} = 0.0010010 > 0.001$ $n = 499, \frac{1}{2n+3} = 0.00099900 \leq 0.001$ $\therefore n = 499.$
--	--

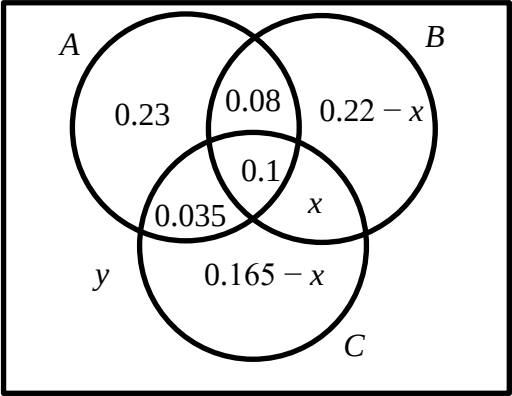
Qn 5	Suggested Solutions [Out of syllabus (9758)]
-------------	---

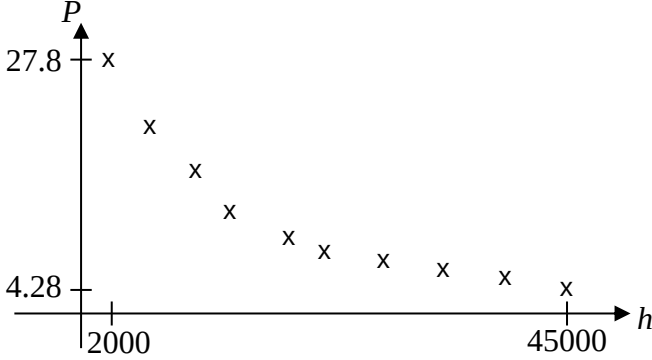
Qn 6	Suggested Solutions
(i)	Let X be the number of sweets, out of 10, that is red in colour. $X \sim B(10, 0.25)$ $P(X \geq 4) = 1 - P(X \leq 3)$ $= 0.22412$ ≈ 0.224 (correct to 3 s.f)
(ii)	[Out of syllabus (9758)]
(iii)	[Out of syllabus (9758)]

Qn 7	Suggested Solutions [Out of syllabus (9758)]
-------------	---

Qn 8	Suggested Solutions [Out of syllabus (9758)]
-------------	---

Qn 9	Suggested Solutions
(i)	$P(B A) = P(B) = 0.4$, since A and B are independent.
(ii)	$P(A \cap B) = P(A) \times P(B) = 0.18$, since A and B are independent. $P(A \cap C) = P(A) \times P(C) = 0.135$, since A and C are independent. $P(B \cap C) = P(B) \times P(C) = 0.12$, since B and C are independent. Representing values into Venn diagram,  Presenting values on a Venn diagram is important to solve this question. From Venn diagram, $P(A' \cap B' \cap C') = 1 - 0.235 - 0.035 - 0.145$ $- 0.08 - 0.1 - 0.02 - 0.2$ $= 0.185.$

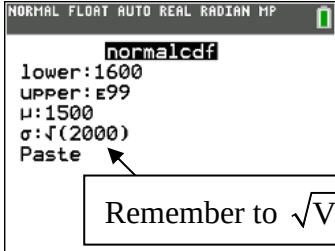
(iii)	Let $P(B \cap C \cap A') = x$. Let $P(A' \cap B' \cap C') = y$.  From Venn diagram, $\therefore 0.235 + 0.08 + 0.035 + 0.1 + x + 0.22 - x + 0.165 - x + y = 1$ $0.835 - x + y = 1$ $y = 0.165 + x$ For minimum y, let $x = 0 \Rightarrow y = 0.165$ For maximum y, maximum $x = 0.165 \Rightarrow y = 0.165 + 0.165 = 0.33$ $\therefore$ least $P(A' \cap B' \cap C') = 0.165$, greatest $P(A' \cap B' \cap C') = 0.33$.
---------------------	---

10	Suggested Solutions
(i)	
(ii)(a)	$r = -0.98073 \approx -0.9807$ (correct to 4 d.p)
(ii)(b)	$r = -0.97480 \approx -0.9748$ (correct to 4 d.p)
(ii)(c)	$r = -0.99864 \approx -0.9986$ (correct to 4 d.p)
(iii)	Since for part (c), $r = -0.99864$ is closest to -1, and as h increases, P decreases at a decreasing rate, part (c) is the most appropriate model. $\therefore P = 34.789 - 0.14687\sqrt{h}$ $P = 34.8 - 0.147\sqrt{h} \text{ (correct to 3 s.f)}$

(iv)	Since 1 metre = 3.28 feet, x metre $\Rightarrow 3.28x$ feet $P = 34.789 - 0.14687\sqrt{3.28x}$ $P = 34.789 - 0.26599\sqrt{x}$ $P = 34.789 - 0.26599\sqrt{h}$ (h is in metres) $P = 34.8 - 0.266\sqrt{h}$ (correct to 3 s.f) (h is in metres) Otherwise method: Convert all data values to meters and recompute. You will get the same answer. For higher ability students you can show them that the r values computed for data values in feet vs in meters are the same (independent of scale of measurement).	Given height in metres (x metre), convert into feet and substitute into equation. Since equation is in feet you substitute $3.28x$ feet into the equation.
------	--	--

Qn 11	Suggested Solutions	
(i)	CABGES AB Number of different arrangements = $\frac{8!}{2!2!} = 10080$.	8 alphabets 2 repeated 'A' 2 repeated 'B'
(ii)	(There is only 1 arrangement that is in alphabetical order, i.e.: AABBBCEGS) $\therefore$ Number of different arrangements = $10080 - 1 = 10079$.	
(iii)	By grouping, <u>AA</u> <u>BB</u> <u>C</u> <u>G</u> <u>E</u> <u>S</u> $\therefore$ Number of different arrangements = $6! = 720$.	No need to permute within 'AA' as both A are identical. Permute 6 groups
(iv)	Part (iii) Case 1: 'A's together <u>AA</u> <u>B</u> <u>B</u> <u>C</u> <u>G</u> <u>E</u> <u>S</u> $\therefore$ Number of different arrangements = $\frac{7!}{2!} = 2520$. Case 2: 'B's together <u>BB</u> <u>A</u> <u>A</u> <u>C</u> <u>G</u> <u>E</u> <u>S</u> $\therefore$ Number of different arrangements = $\frac{7!}{2!} = 2520$.	

	$\therefore$ Total number of different arrangements $= \text{Total} - [\text{Case 1} + \text{Case 2} - \text{part(iii)}]$ $= 10080 - (2520 + 2520 - 720)$ $= 5760.$
--	--

Qn 12	Suggested Solutions
(i)	Let X be the mass of a randomly chosen apple in grams. Let Y be the mass of a randomly chosen pear in grams. $X \sim N(300, 20^2)$, $Y \sim N(200, 15^2)$ Let $W = X_1 + X_2 + X_3 + X_4 + X_5$. $E(W) = 5 \times 300 = 1500$ $\text{Var}(W) = 5 \times 20^2 = 2000$ $\therefore W \sim N(1500, 2000)$ $P(W > 1600) = 0.012674$ ≈ 0.0127 (correct to 3 s.f)  Remember to $\sqrt{\text{Variance}}$
(ii)	<u>Aim:</u> $P(X_1 + X_2 + X_3 + X_4 + X_5 > Y_1 + Y_2 + \dots + Y_8)$ $= P[(X_1 + X_2 + X_3 + X_4 + X_5) - (Y_1 + Y_2 + \dots + Y_8) > 0]$ Let $A = (X_1 + X_2 + X_3 + X_4 + X_5) - (Y_1 + Y_2 + \dots + Y_8)$ $E(A) = 5E(X) - 8E(Y) = 5 \times 300 - 8 \times 200 = -100$ $\text{Var}(A) = \text{Var}(X_1 + X_2 + X_3 + X_4 + X_5) + \text{Var}(Y_1 + Y_2 + \dots + Y_8)$ $= 5\text{Var}(X) + 8\text{Var}(Y) = 5 \times 20^2 + 8 \times 15^2 = 3800$ $\therefore A \sim N(-100, 3800)$ $P(A > 0) = 0.052379$ ≈ 0.0524 (correct to 3 s.f)
(iii)	<u>Aim:</u> $P[0.85(X_1 + X_2 + X_3 + X_4 + X_5) + 0.9(Y_1 + Y_2 + \dots + Y_8) < 2750]$ Let $C = 0.85(X_1 + X_2 + X_3 + X_4 + X_5) + 0.9(Y_1 + Y_2 + \dots + Y_8)$ $E(C) = 0.85 \times E(X_1 + X_2 + X_3 + X_4 + X_5) + 0.9 \times E(Y_1 + Y_2 + \dots + Y_8)$ $= 0.85[5E(X)] + 0.9[8E(Y)]$ $= 0.85 \times 5 \times 300 + 0.9 \times 8 \times 200$ $= 2715$ $\text{Var}(C) = 0.85^2 [\text{Var}(X_1 + X_2 + X_3 + X_4 + X_5)] + 0.9^2 [\text{Var}(Y_1 + Y_2 + \dots + Y_8)]$ $= 0.85^2 [5\text{Var}(X)] + 0.9^2 [8\text{Var}(Y)]$ $= 0.85^2 \times 5 \times 20^2 + 0.9^2 \times 8 \times 15^2$ $= 2903$ $\therefore C \sim N(2715, 2903)$ $\text{Var}(aX) = a^2 \text{Var}(X)$ $\text{Var}(X_1 + X_2) = 2\text{Var}(X)$

	$P(C < 2750) = 0.74202$ ≈ 0.742 (correct to 3 s.f)
--	---